STEEL BEAM / 01

Beam analysis.

Understand how your beam responds before the work begins.

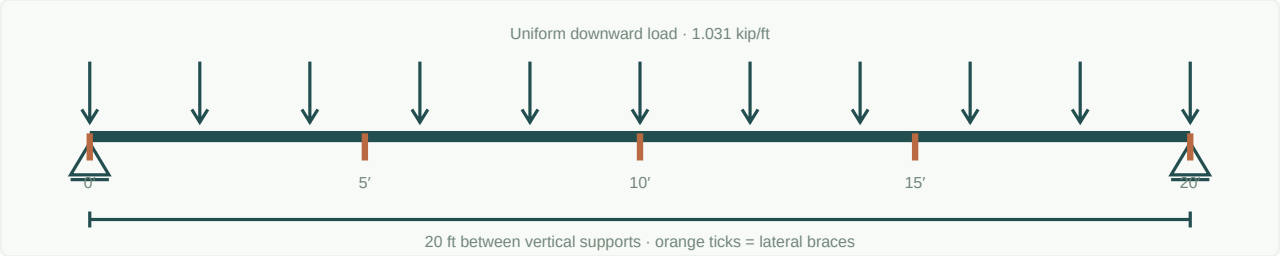
> **For preliminary evaluation only.** This analysis is not a construction approval or a complete member design. Connections, bearing, stability and scaffold-specific requirements require separate review.

02 / VISUALIZE

SCHEMATIC

Beam arrangement

A clear view of the model being analyzed.



BEAM ↓ APPLIED LOAD

NOT TO SCALE

03 / REVIEW

PRELIMINARY

Analysis results

Responses for the load case shown above.

Support reaction · each **10.31** kip	Maximum moment **51.55** kip-ft
Nominal EI deflection **0.3413** in	Reduced EI deflection **0.4266** in

Calculation basis

Section W16X31 · Span 20 ft · E = 29,000 ksi · Ix = 375 in⁴. Nominal section properties and weight are from a third-party mirror of AISC Shapes Database v15; verify against the official source. The inputs are user-selected and the beam is simply supported with full-span uniform downward loading.

1.0D + 1.0L: 0.2 + 0.8 + 0.031 self-weight = **1.031 kip/ft**. Self-weight is included once.

R = wL/2 = 1.031 × 20 / 2 = 10.31 kip. M = wL²/8 = 51.55 kip-ft. Maximum bending deflection = 5wL⁴/(384EI), at 10 ft. The reduced value uses 0.8EI; the nominal value uses EI.

Lateral brace locations: 0, 5, 10, 15, 20 ft. Segment lengths: 5, 5, 5, 5 ft. Braces are not vertical supports.

Additional information required. Bending, shear, stability, bearing, connections and scaffold-specific capacity are not evaluated. No overall member adequacy is determined. Independent comparison not completed. The stiffness reduction is a selected standalone assumption, not a complete Direct Analysis Method. Shear deformation is excluded.

Project: Untitled project · Prepared by: Not specified · 2026-11-09T21:35:32.514Z · Engine 0.1.0

ⓘ **Independent comparison not completed**
Member capacity and scaffold-specific checks are not evaluated by this analysis.

04 / EXPLORE

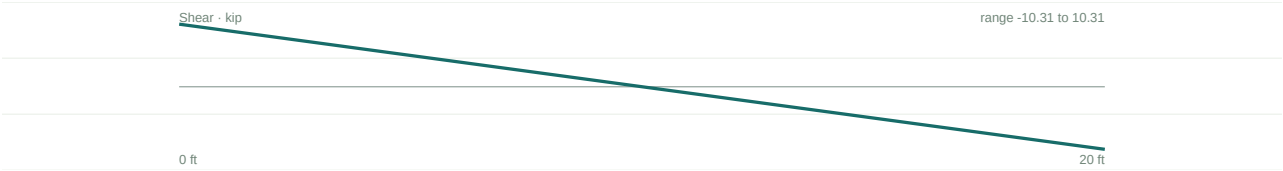
SERVICE LOAD

Response diagrams

View how forces and displacement change along the span.

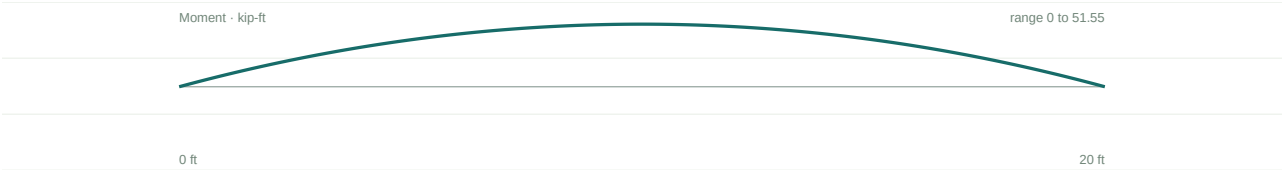
Shear force

kips



Bending moment

kip-ft



Deflection

in



05 / REFERENCE

AISC v15 · mirrored

Section properties

AISC v15 properties from a third-party mirror; verify against the official source before use.

W **31** lb/ft	A **9.13** in^2	d **15.9** in
bf **5.53** in	tw **0.275** in	tf **0.44** in
Ix **375** in^4	Sx **47.2** in^3	Zx **54** in^3
Iy **12.4** in^4	Sy **4.49** in^3	Zy **7.03** in^3
J **0.461** in^4	Cw **739** in^6	rx **6.41** in
ry **1.17** in	rts **1.42** in	ho **15.5** in

Terms of use · draft

This tool supports preliminary planning and does not replace review by a qualified professional. Verify inputs, applicable standards, connections, bearing, stability, and all scaffold-specific requirements before use. No certification or construction approval is provided.